

**YEAR 9 SCIENCE
CHEMICAL SCIENCES
TEST 1 - ATOMIC STRUCTURE, RADIATION**

NAME: _____

CLASS: _____

Mark: 26

All matter is made of atoms that are composed of protons, neutrons and electrons; natural radioactivity arises from the decay of nuclei in atoms.

Multiple Choice Write the answer to each question in the appropriate box at right.

1. γ , β , α , B

IN ORDER, the above symbols mean:

- A Gamma, Beta, Alpha, Boron.
- B Gamma, Boron, Alpha, Beta.
- C Alpha, Beta, Gamma, Beryllium.
- D Alpha, Beryllium, Gamma, Beta.

2. The nucleus of which atom is identical to an **alpha particle**?

- A H - Hydrogen
- B He - Helium
- C Li - Lithium
- D Ti - Titanium

3. Which of the following statements about atoms is **true**?

- A An atom is mostly empty space.
- B Neutrons and protons are found in the nucleus with the electrons.
- C Electrons circle the nucleus all the time.
- D The number of neutrons found in the nucleus of an atom is always the same.

4. The **mass number** of an element is the total number of:

- A electrons in an atom.
- B protons in an atom.
- C protons and electrons in an atom.
- D protons and neutrons in an atom.

5. The **atomic number** of an element is the total number of:

- A electrons in an atom.
- B protons in an atom.
- C protons and electrons in an atom.
- D protons and neutrons in an atom.

QUESTION	ANSWER
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6. An atom contains 4 protons, 5 neutrons and 4 electrons. Its mass number is:
- A 4
 - B 5
 - C 9
 - D 13
7. An atom contains 4 protons, 5 neutrons and 4 electrons. Its atomic number is:
- A 4
 - B 5
 - C 9
 - D 13
8. A charged atom is called:
- A an isotope.
 - B an ion.
 - C a proton.
 - D radioactive.
9. Define ionising radiation:
- A Radiation from the ionosphere.
 - B Radiation produced when atoms are ionised.
 - C Any type of electromagnetic radiation.
 - D Radiation that can remove electrons from atoms and molecules.
10. Identify the set that contains an element, a compound and a mixture.
- A Carbon dioxide, air, silver.
 - B Gold, sulfur, phosphorus.
 - C Granite, oxygen, hydrogen.
 - D Copper oxide, calcium carbonate, sodium chloride.
11. Beta particles are:
- A fast-moving electrons.
 - B positively charged.
 - C heavier than an alpha particle.
 - D unable to penetrate human skin.
12. One of the particles inside the atom is very much lighter than the others. It is the:
- A electron.
 - B neutron.
 - C nucleus.
 - D proton.
13. The nucleus of the atom contains:
- A only neutrons.
 - B only protons.
 - C electrons and neutrons.
 - D neutrons and protons.

14. Which of the following are atoms of the same element?

	Number of protons	Number of neutrons	Number of electrons
Atom U	11	13	11
Atom V	13	11	13
Atom X	12	11	13
Atom Y	11	12	11

- A U and Y
- B U and X
- C X and Y
- D U and V

15. Different isotopes of the same element contain different numbers of:

- A electrons.
- B neutrons.
- C protons.
- D nuclei.

16. Iodine -131 has a half-life of 8 days. If a sample of I-131 has a mass of 60 g, how much radioactive iodine will be left after 16 days?

- (a) 7.5 g
- (b) 15 g
- (c) 30 g
- (d) 3.75 g

Short Answer Write the answer to the questions in the spaces provided.

1. Complete the following table.

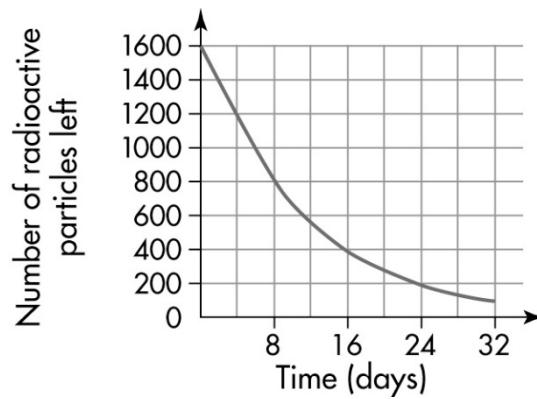
ELEMENT	PROTONS	NEUTRONS	ELECTRONS
$^{19}_{9}\text{F}$			
$^{56}_{26}\text{Fe}$			
$^{39}_{19}\text{K}$			

(3)

2. What makes the nuclei of some atoms naturally unstable?

(1)

3. The graph below shows the radioactive decay of iodine-131.



- (a) What is the half-life of iodine-131?

(1)

- (b) How many radioactive particles would be left after 40 days?

(1)

4. List **two** major long-term effects on humans caused by cell damage resulting from exposure to large amounts of nuclear radiation.

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(2)

5. Describe **two** uses of nuclear radiation in medicine and/or industry.

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(2)